NUMBER & ALGEBRA

Chapter 18

Complex Numbers

18

SYLLABUS 1.12–1.14 HL ONLY

In the sixteenth century, Italian mathematicians learned to solve cubic equations. Their formula produced something unexpected. It sometimes asked for the square root of a negative number, even when the equation had three perfectly ordinary answers. No number they knew could give a negative square. Yet if they wrote the impossible quantity down and kept calculating, it disappeared before the end, and the answers were correct.

This chapter begins with that impossible quantity. We define a number i with $i^2 = -1$. No real number does this, so i is new. For two centuries mathematicians did not trust it, and the name "imaginary" was chosen as an insult. The name was unfair. Once i is allowed, every polynomial equation has a solution, a result now called the **Fundamental Theorem of Algebra**.

These new numbers are also geometric. Each one is a point in a plane, with the real numbers along one axis. Adding two of them moves a point across the plane. Multiplying two of them rotates and stretches. Multiplying by i rotates by a quarter-turn. In ordinary coordinates a rotation takes several steps; here it is one multiplication. This is why complex numbers are used throughout physics and engineering.

By the end of this chapter you will be able to write a complex number in three forms: $a + bi$, $r\operatorname{cis}\theta$ and $re^{i\theta}$. You will multiply, divide, and find powers and roots in each form. You will also use complex roots to factorise polynomials and to prove trigonometric identities.

Three forms may seem like more than you need. Each one makes a different calculation easy. Adding is simplest in $a + bi$ form. A power that would fill a page of working takes one line in polar form. Choosing the form before you start is the main skill in this chapter.

We start with that algebra: numbers of the form $a + bi$.

18.1 Cartesian Form

Everything starts from a single definition that no real number satisfies.

DEF · IB The **imaginary unit** i satisfies $i^2 = -1$. A **complex number** is any $z = a + bi$ with $a, b \in \mathbb{R}$; $a = \text{Re}(z)$ is the **real part** and $b = \text{Im}(z)$ the **imaginary part**.

You add, subtract and multiply complex numbers just as you would binomials in i, with the single extra rule that i^2 is replaced by -1 wherever it appears.

That one rule is enough to find every power of i, because the powers repeat in a cycle of four:

$$i^1 = i, \qquad i^2 = -1, \qquad i^3 = i^2 \cdot i = -i, \qquad i^4 = (i^2)^2 = 1,$$

and $i^5 = i^4 \cdot i = i$ starts the cycle again. To simplify any power, divide the exponent by 4 and keep only the remainder: $i^{2026} = i^{4(506)+2} = i^2 = -1$.

EXAMPLE 18.1

Given $z = 2 + i$ and $w = 5 - 3i$, find $z + w$ and zw.

Adding is done part by part: $z+w = (2+5)+(1-3)i = 7-2i$. Multiplying, expand and then use $i^2 = -1$:

$$zw = (2 + i)(5 - 3i) = 10 - 6i + 5i - 3i^2 = 10 - i + 3 = 13 - i.$$

The whole of complex arithmetic is ordinary algebra plus the one substitution $i^2 = -1$.

The conjugate and division

Every complex number has a mirror image across the real axis, called its **conjugate**. The conjugate is what makes division possible.

DEF The **complex conjugate** of $z = a + bi$ is $z^* = a - bi$.

The conjugate matters because a number times its own conjugate is always real: $zz^* = (a + bi)(a - bi) = a^2 + b^2$. This is exactly the tool for division, in the same way that a surd is rationalised. To divide, multiply top and bottom by the conjugate of the denominator, turning the denominator real.

EXAMPLE 18.2

Find $\dfrac{3 + 2i}{1 - i}$.

Multiply numerator and denominator by the conjugate $1+i$:

$$\frac{3+2i}{1-i}\cdot\frac{1+i}{1+i}=\frac{3+3i+2i+2i^2}{1^2+1^2}=\frac{3+5i-2}{2}=\frac{1+5i}{2}=\tfrac{1}{2}+\tfrac{5}{2}i.$$

Multiplying by the conjugate removes i from the denominator every time; the answer is then read off as real and imaginary parts.

One more Cartesian tool: two complex numbers are equal exactly when their real parts agree *and* their imaginary parts agree. A single complex equation therefore carries two real equations at once, and **equating parts** solves for unknowns.

EXAMPLE 18.3

Find the complex number z satisfying $z+2z^*=9+2i$.
Write $z=a+bi$, so $z^*=a-bi$:

$$(a+bi)+2(a-bi)=3a-bi=9+2i.$$

Equate real parts: $3a=9 \implies a=3$. Equate imaginary parts: $-b=2 \implies b=-2$. So $z=3-2i$. One equation was enough for two unknowns, because the real and imaginary parts each give an equation.

The Argand diagram

A complex number carries two real numbers, so it is naturally a *point* in a plane: the horizontal axis for the real part, the vertical for the imaginary. This picture, the **Argand diagram**, is what makes the whole subject geometric.

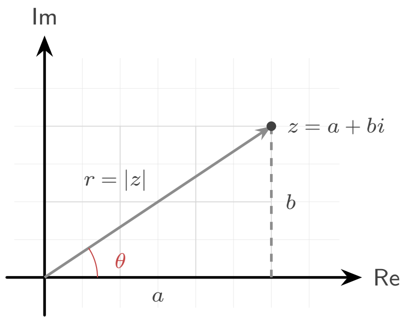

Figure 18.1 A complex number $z=a+bi$ drawn as the point (a,b) of the Argand diagram. Its distance from the origin is the modulus r, and the angle its arrow makes with the positive real axis is the argument θ.

The picture immediately explains the arithmetic. **Addition is vector addition**: to add w to z, place w's arrow tail-to-tip on z's, exactly as in Ch. 10, so $z+w$ is the far corner of the parallelogram the two arrows span. And **conjugation is reflection**: z^* is z flipped across the real axis, which is why zz^* lies on the real axis.

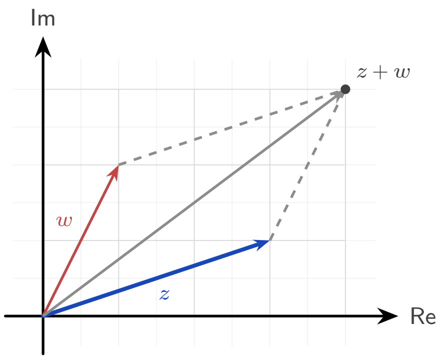

Figure 18.2 Addition is vector addition. Placing w's arrow tail-to-tip on z's puts $z+w$ at the far corner of the parallelogram the two arrows span.

There are two natural numbers to attach to a point in a plane: how far it is from the origin, and in what direction. These are the modulus and the argument, and they are the subject of the next section.

YOUR TURN 18.1 Cartesian Form

1. Evaluate each expression, giving your answer in the form $a+bi$.

(a) $(3+2i)+(1-4i)$ [1]
(b) $(5-i)-(2+3i)$ [1]
(c) $(2+3i)(1-i)$ [2]
(d) $(2+i)(2-i)$ [1]

2. Simplify each power of i.

(a) i^3 [1]
(b) i^4 [1]
(c) i^{15} [2]
(d) i^{100} [2]

3. Let $z=7-4i$.

(a) Write down Re(z) and Im(z). [1]
(b) Write down z^*. [1]
(c) Evaluate zz^*. [2]

4. Write each quotient in the form $a+bi$.

(a) $\dfrac{4+3i}{2-i}$ [3]
(b) $\dfrac{1}{1+i}$ [2]

5. Find the complex number z in each case.

(a) $z + 3z^* = 8 - 4i$ [3]

(b) $2z - z^* = 3 + 6i$ [3]

18.2 Modulus and Argument

The distance of z from the origin is its **modulus** $|z|$, straight from Pythagoras; the angle its arrow makes with the positive real axis is its **argument** $\arg(z)$, measured anticlockwise.

KEY For $z = a + bi$: $|z| = \sqrt{a^2 + b^2}$, $\tan(\arg z) = \dfrac{b}{a}$.

CAUTION *Finding the argument needs care:* $\tan\theta = \frac{b}{a}$ *alone cannot tell which quadrant z is in, because* tan *repeats every π. Always sketch z on the Argand diagram first, then choose the angle in the correct quadrant, conventionally in the range $-\pi < \theta \le \pi$ (the* ***principal argument****).*

Knowing the modulus r and argument θ is enough to rebuild the number, since $a = r\cos\theta$ and $b = r\sin\theta$. This gives the **modulus–argument** (or polar) form, and the equivalent exponential version, **Euler's form**.

KEY · IB **Polar and Euler form:** $z = r(\cos\theta + i\sin\theta) = r\,\mathrm{cis}\,\theta = re^{i\theta}$, where $r = |z|$ and $\theta = \arg(z)$.

That $e^{i\theta} = \cos\theta + i\sin\theta$ is startling on first sight, and you already have the tools to see why it is true. Take the Maclaurin series for e^x from Ch. 17 and substitute $x = i\theta$, using the cycle of powers of i from §18.1:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \cdots\right).$$

The even terms, where i has become ± 1, form the cosine series; the odd terms, which still carry an i, form the sine series. The exponential function and the trigonometric functions are two views of one function on the complex plane. Setting $\theta = \pi$ gives the famous $e^{i\pi} + 1 = 0$, which ties e, i, π, 1 and 0 together in a single line, and now you know it is not a coincidence but a calculation.

EXAMPLE 18.4

Write $z = 1 - i$ in polar and Euler form, with $-\pi < \theta \le \pi$.

The modulus is $r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$. The point $(1, -1)$ lies in the fourth quadrant, so the argument is $\theta = -\frac{\pi}{4}$. Hence

$$z = \sqrt{2}\,(\cos \tfrac{-\pi}{4} + i \sin \tfrac{-\pi}{4}) = \sqrt{2}\ \text{cis}(-\tfrac{\pi}{4}) = \sqrt{2}\,e^{-i\pi/4}.$$

Sketch first, then read off the quadrant. A common error is to take arctan without checking which quadrant the point is in.

YOUR TURN 18.2 Modulus and Argument

6. Find the modulus of each number.

(a) $3 + 4i$ [1]
(b) $5 - 12i$ [1]
(c) $1 + i$ [1]
(d) $-2i$ [1]

7. Find the principal argument of each number, with $-\pi < \theta \leq \pi$.

(a) $1 + i$ [1]
(b) i [1]
(c) -3 [1]
(d) $-1 - i$ [2]

8. Write each number in modulus–argument form and in Euler form.

(a) $2i$ [2]
(b) -3 [2]
(c) $1 - i$ [3]

9. Write each number in Cartesian form.

(a) $4\,\text{cis}\,\frac{\pi}{3}$ [2]
(b) $2e^{-i\pi/4}$ [2]

18.3 Products, Quotients and De Moivre's Theorem

Polar form is worth the extra notation because it makes multiplication easy. When two complex numbers are multiplied, their moduli multiply and their arguments *add*, which is just the law of exponents applied to $re^{i\theta}$.

KEY $|zw| = |z|\,|w|$ and $\arg(zw) = \arg z + \arg w$; $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$ and $\arg\left(\frac{z}{w}\right) = \arg z - \arg w$.

Geometrically this is striking: multiplying by a complex number *rotates* by its argument and *scales* by its modulus. So multiplying by $i = 1 \cdot e^{i\pi/2}$, whose modulus is 1, is a quarter-turn anticlockwise with no stretching at all.

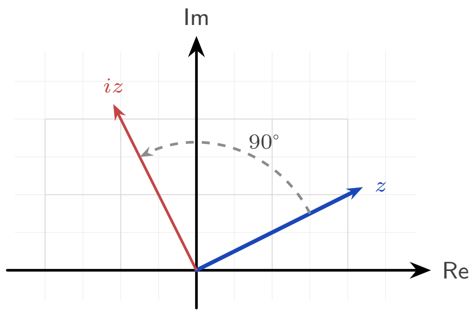

Figure 18.3 Multiplying by i turns a complex number through a quarter-turn anticlockwise about the origin, leaving its modulus unchanged.

If multiplying adds arguments, then raising to a power *multiplies* the argument. This is **De Moivre's theorem**, and everything in the rest of the chapter follows from it.

KEY · IB **De Moivre's theorem:** $\left(r(\cos\theta + i\sin\theta)\right)^n = r^n(\cos n\theta + i\sin n\theta) = r^n e^{in\theta}$, for $n \in \mathbb{Z}$.

The syllabus extends the theorem to *rational* exponents (indeed it holds for all real n): a power like $z^{1/n}$ obeys the same rule, though it becomes multi-valued, which is exactly the story of the nth roots in the next section. For positive integers the theorem is proved by the induction of Ch. 3, using the compound-angle identities of Ch. 9.

EXAMPLE 18.5

Prove De Moivre's theorem for $n \in \mathbb{Z}^+$ by mathematical induction.

Let P_n be the statement $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

Show P_1 is true: the left side is $(\cos\theta + i\sin\theta)^1$, the right is $\cos\theta + i\sin\theta$. Equal.

Assume P_k is true for some $k \in \mathbb{Z}^+$. Then

$$(\cos\theta + i\sin\theta)^{k+1} = (\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)$$

$$= (\cos k\theta\cos\theta - \sin k\theta\sin\theta) + i(\sin k\theta\cos\theta + \cos k\theta\sin\theta) = \cos(k+1)\theta + i\sin(k+1)\theta,$$

by the compound-angle identities. So P_k implies P_{k+1}.

Hence, since P_1 is true and $P_k \implies P_{k+1}$, the statement P_n is true for all $n \in \mathbb{Z}^+$ by mathematical induction. ■

> Notice what the proof shows: each step of De Moivre is one use of the compound-angle identities. The theorem is those identities of Ch. 9 applied n times.

Raising a complex number to a high power means expanding a bracket again and again in Cartesian form. In polar form it takes one line: raise the modulus to the power, multiply the angle by the power, done.

> **EXAMPLE 18.6**
>
> Find $(1+i)^8$.
>
> In polar form $1+i = \sqrt{2}\ \text{cis}\ \frac{\pi}{4}$. By De Moivre,
>
> $$(1+i)^8 = (\sqrt{2})^8\ \text{cis}(8 \cdot \tfrac{\pi}{4}) = 16\ \text{cis}(2\pi) = 16.$$
>
> Eight multiplications reduce to one power of the modulus and one multiple of the angle.

Draw the powers and you can watch the theorem work. Each multiplication by $1+i$ turns the arrow 45° anticlockwise and stretches it by $\sqrt{2}$, so the powers lie on a spiral that winds outward from the origin:

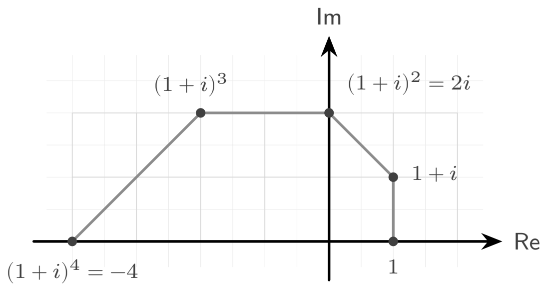

Figure 18.4 The first five powers of $1+i$. Each multiplication turns the arrow through 45° and stretches it by $\sqrt{2}$, so the powers lie on a spiral winding outward from the origin.

Four more steps continue the spiral through the lower half-plane and back to the positive real axis at $(1+i)^8 = 16$: a full turn of $8 \times 45^\circ = 360^\circ$, with the modulus grown to $(\sqrt{2})^8 = 16$. The worked example above is this picture written in algebra.

Numbers on the unit circle

One case of De Moivre is worth stating separately, because it is used often. If $|z| = 1$ then $z = \text{cis}\,\theta$, and dividing 1 by a number of modulus 1 simply reverses its angle, so $\dfrac{1}{z} = \text{cis}(-\theta) = z^*$. Add the two and the imaginary parts cancel while the real parts double; subtract them and the opposite happens.

KEY For $z = \text{cis}\,\theta$ and any $n \in \mathbb{Z}$:

$$z^n + \frac{1}{z^n} = 2\cos n\theta, \qquad z^n - \frac{1}{z^n} = 2i\sin n\theta.$$

Read from left to right this converts powers into multiple angles, and that is what makes it useful: an expression like $\cos^4\theta$, awkward to integrate as it stands, becomes a sum of $\cos 2\theta$ and $\cos 4\theta$ terms that can be integrated directly. You will build exactly that in the end-of-chapter problems.

YOUR TURN 18.3 Products, Quotients and De Moivre's Theorem

10. Evaluate each expression, giving your answer in modulus–argument form.

(a) $\left(2\,\text{cis}\,\frac{\pi}{6}\right)\left(3\,\text{cis}\,\frac{\pi}{3}\right)$ [2]

(b) $\dfrac{6\,\text{cis}\,\frac{2\pi}{3}}{2\,\text{cis}\,\frac{\pi}{3}}$ [2]

(c) $\left(\text{cis}\,\frac{\pi}{5}\right)^5$ [2]

(d) $\left(2\,\text{cis}\,\frac{\pi}{6}\right)^3$ [2]

11. Given $|z| = 3$, $|w| = 4$, $\arg z = \frac{\pi}{4}$ and $\arg w = \frac{\pi}{6}$, write down

(a) $|zw|$ [1]

(b) $\arg(zw)$ [1]

(c) $\left|\dfrac{z}{w}\right|$ [1]

(d) $\arg\left(\dfrac{z}{w}\right)$ [1]

12.

(a) Expand $(\cos\theta + i\sin\theta)^4$ using De Moivre's theorem. [1]

(b) Use De Moivre's theorem to evaluate $(1+i)^6$. [3]

13. Let $z = \text{cis}\,\theta$. Write each expression in terms of a single cosine or sine.

(a) $z + \dfrac{1}{z}$ [1]

(b) $z^3 + \dfrac{1}{z^3}$ [2]

(c) $z^2 - \dfrac{1}{z^2}$ [2]

18.4 Roots of Complex Numbers

De Moivre also works in reverse. Just as a positive real number has two square roots, a complex number has exactly n distinct nth roots, and they are evenly spaced around a circle. To find them, write the number in polar form, take the nth root of the modulus, divide the argument by n, and then step around by $\frac{2\pi}{n}$ to collect all n of them.

KEY The nth roots of $z = r\,\mathrm{cis}\,\theta$ are $\sqrt[n]{r}\,\mathrm{cis}\left(\dfrac{\theta + 2\pi k}{n}\right)$, for $k = 0, 1, \ldots, n-1$.

Because the roots share a modulus $\sqrt[n]{r}$ and their arguments differ by equal steps of $\frac{2\pi}{n}$, they sit at the vertices of a **regular n-gon** centred at the origin. The special case $z^n = 1$ gives the **roots of unity**, a regular n-gon with one vertex at 1.

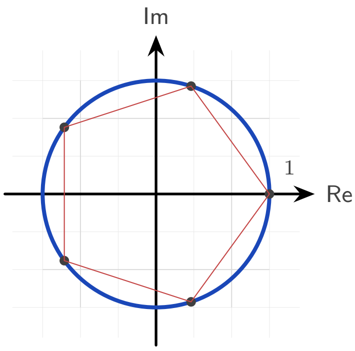

Figure 18.5 The five fifth roots of unity, at the vertices of a regular pentagon inscribed in the unit circle with one vertex at 1.

The roots of unity are worth a closer look, because the whole set is generated by just one of them. Write $\omega = \mathrm{cis}\,\dfrac{2\pi}{n}$ for the first root anticlockwise from 1. Then $\omega^k = \mathrm{cis}\,\dfrac{2\pi k}{n}$ is the kth root round the circle, and $\left(\omega^k\right)^n = \left(\omega^n\right)^k = 1^k = 1$, so every power of ω is again a root. The n roots are therefore

$$1,\ \omega,\ \omega^2,\ \ldots,\ \omega^{n-1},$$

one number and its powers, stepping round the polygon and returning to the start.

They also **sum to zero**. Geometrically the n arrows point symmetrically in every direction and cancel, but the geometric series of Ch. 6 proves it outright: since $\omega \neq 1$,

$$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega - 1} = 0.$$

The same conclusion follows from Ch. 7: $z^n - 1$ has no z^{n-1} term, so the sum of its roots vanishes. For $n = 3$ this gives the identity $1 + \omega + \omega^2 = 0$, which appears whenever cube roots of unity are used.

These evenly spaced points are also the basis of the *discrete Fourier transform*, which separates a signal into pure frequencies by sampling it against precisely these n points on the circle. The fast algorithm for computing it, the FFT, uses the polygon's symmetry, and it is what makes audio compression and digital signal processing fast enough to run in real time on a phone.

EXAMPLE 18.7

Find the three cube roots of 8.
Write $8 = 8\,\text{cis}\,0$. The modulus of each root is $\sqrt[3]{8} = 2$, and the arguments are $\frac{0+2\pi k}{3}$ for $k = 0, 1, 2$:

$$2\,\text{cis}\,0 = 2, \qquad 2\,\text{cis}\,\tfrac{2\pi}{3} = -1 + \sqrt{3}\,i, \qquad 2\,\text{cis}\,\tfrac{4\pi}{3} = -1 - \sqrt{3}\,i.$$

The three roots form an equilateral triangle on the circle of radius 2, one real and a conjugate pair.

Square roots without polar form

Polar form gives roots of every order, but it has one drawback: it needs the argument, and unless the angle is one of the standard ones, arctan returns a decimal and the answer is no longer exact. For *square* roots there is a second route that stays in Cartesian form and keeps the surds intact. Assume the root is $a + bi$, square it, and equate real and imaginary parts as in §18.1.

EXAMPLE 18.8

Find the square roots of $3 + 4i$ in the form $a + bi$.
Let $(a + bi)^2 = 3 + 4i$. Expanding the left side gives $a^2 - b^2 + 2abi$, so equating parts,

$$a^2 - b^2 = 3 \qquad \text{and} \qquad 2ab = 4.$$

The second gives $b = \dfrac{2}{a}$. Substituting into the first,

$$a^2 - \frac{4}{a^2} = 3 \implies a^4 - 3a^2 - 4 = 0 \implies (a^2 - 4)(a^2 + 1) = 0.$$

Since a is real, $a^2 = 4$, so $a = \pm 2$ and correspondingly $b = \pm 1$. The square roots are $2 + i$ and $-2 - i$, that is $\pm(2 + i)$.

Two features of that calculation are worth noting. The substitution always produces a **quadratic in a^2**, and the negative solution for a^2 is always rejected, because a is real. The two answers always differ by a sign, since square roots lie at opposite ends of a diameter, half a turn apart.

CAUTION *Use Cartesian equating for square roots when the answer should be exact. For cube roots and higher, go back to polar form: equating parts would give a cubic or an equation of even higher degree, while De Moivre gives roots of any order in one line.*

YOUR TURN 18.4 Roots of Complex Numbers

14.

(a) Write down a square root of -9. [1]
(b) State the modulus of each cube root of 27. [1]
(c) State how many distinct fifth roots a nonzero complex number has. [1]
(d) State the shape formed by the nth roots of a complex number on the Argand diagram. [1]

15. State

(a) the three cube roots of unity; [2]
(b) the four fourth roots of unity. [2]

16. Find the square roots of each number in the form $a+bi$ by equating real and imaginary parts.

(a) $5+12i$ [3]
(b) $-8+6i$ [3]

17. Find, in modulus–argument form,

(a) the three cube roots of 8; [3]
(b) the four fourth roots of 16. [3]

18.5 Complex Roots of Polynomials

Complex numbers were invented to solve equations, and they solve *every* polynomial equation. This is the **Fundamental Theorem of Algebra**: every polynomial of degree $n \geq 1$ has exactly n roots in the complex numbers, counted with multiplicity. The roots of a polynomial can all lie off the real line, but none of them can lie outside the complex plane.

The simplest case is one you have met before in a different form. In Ch. 4, a quadratic with negative discriminant had "no solutions." It always had two roots; they were simply not on the real line.

EXAMPLE 18.9

Solve $z^2 - 4z + 13 = 0$.

The quadratic formula works unchanged:

$$z = \frac{4 \pm \sqrt{16-52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i.$$

The negative discriminant, which once ended the question, now simply produces an i: the two roots are $2+3i$ and $2-3i$, a conjugate pair placed symmetrically above and below the real axis.

That symmetry is no accident. When a polynomial has *real* coefficients, its non-real roots always come in conjugate pairs.

KEY **Conjugate root theorem.** If a polynomial has real coefficients and $z = a + bi$ is a root, then $z^* = a - bi$ is also a root.

This is why a cubic with real coefficients always has at least one real root: its three roots are either all real, or one real root plus a conjugate pair. A non-real root never appears alone in a polynomial with real coefficients.

EXAMPLE 18.10

Given that $2 + i$ is a root of $z^3 - 3z^2 + \cdots$ with real coefficients, state another root and find a quadratic factor.

By the conjugate root theorem, $2 - i$ is also a root. The two together give a quadratic factor with real coefficients:

$$(z - (2+i))(z - (2-i)) = (z-2)^2 - i^2 = z^2 - 4z + 5.$$

A conjugate pair of roots always multiplies to a real quadratic, here z^2-4z+5, which can then be divided out to find the remaining real root.

Trigonometric identities from De Moivre

De Moivre's theorem also produces trigonometric identities. Expand $(\cos\theta + i\sin\theta)^n$ by the binomial theorem of Ch. 2, then compare real and imaginary parts with $\cos n\theta + i\sin n\theta$, and a multiple-angle formula follows for any n.

EXAMPLE 18.11

Use De Moivre's theorem to show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, and find a matching identity for $\sin 3\theta$.

Write $c = \cos\theta$, $s = \sin\theta$, and expand by the binomial theorem:

$$(c+is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = (c^3 - 3cs^2) + i(3c^2s - s^3).$$

By De Moivre this equals $\cos 3\theta + i\sin 3\theta$, so the parts must match separately. Real parts, using $s^2 = 1 - c^2$:

$$\cos 3\theta = c^3 - 3c(1-c^2) = 4\cos^3\theta - 3\cos\theta.$$

Imaginary parts, using $c^2 = 1 - s^2$:

$$\sin 3\theta = 3(1 - s^2)s - s^3 = 3\sin\theta - 4\sin^3\theta.$$

One expansion gives two identities, and the same method works for any n. Results that took several steps with compound angles in Ch. 9 now follow from a single binomial expansion.

TIP *Complex numbers offer several good IA topics. Three directions worth exploring: iterate $z \mapsto z^2 + c$ for different c and investigate which starting points escape to infinity (this is how the Mandelbrot set is defined); investigate which regular polygons can be constructed with ruler and compass, a question Gauss answered at nineteen using exactly the roots of unity above; or measure phase shift in a simple AC circuit and model it with rotating complex numbers.*

YOUR TURN 18.5 Complex Roots of Polynomials

18. Solve each equation.

(a) $z^2 + 9 = 0$ [1]
(b) $z^2 + 4 = 0$ [1]
(c) $z^2 - 2z + 5 = 0$ [2]
(d) $z^2 + 2z + 2 = 0$ [2]

19.

(a) Given that $3+i$ is a root of a polynomial with real coefficients, state another root. [1]
(b) Write down a quadratic with real coefficients whose roots are $2i$ and $-2i$. [2]
(c) Write down a quadratic with real coefficients whose roots are $1 + i$ and $1 - i$. [2]

20.

(a) The equation $z^2 + bz + c = 0$ has real coefficients and $1 + 2i$ as a root. Find b and c. [3]
(b) Explain why a cubic with real coefficients must have at least one real root. [2]

Chapter 18 · Complex Numbers

Reflections

IM The name "imaginary" was an insult. Descartes coined it in the seventeenth century to dismiss these numbers as fictions, and Leibniz called them "an amphibian between being and non-being." Their rehabilitation was slow and international. The Norwegian surveyor Caspar Wessel and the Swiss bookkeeper Jean-Robert Argand independently drew the plane picture that made them concrete. It was the German mathematician Carl Friedrich Gauss who finally made them respectable: he gave the first accepted proof of the Fundamental Theorem of Algebra, and he popularised the term "complex" itself. A number system that began in suspicion became, within two centuries, essential.

TOK Does the naming shape the difficulty? "Imaginary" was coined as a dismissal and "real" as its opposite, and both names outlived the objection by centuries. Would these numbers have been accepted sooner if they had been called something neutral, such as "lateral"? If a change of vocabulary can change how readily an idea is believed, how much of what we call mathematical knowledge is carried by its language rather than its content?

TOK Euler's identity $e^{i\pi} + 1 = 0$ is routinely called the most beautiful equation in mathematics. What is being judged when a mathematician calls a result beautiful, and is that judgement evidence of anything? Beauty in mathematics is often described as economy, surprise, or unexpected connection, all of which this identity has. Yet if beauty guides which results we pursue and teach, it is shaping the subject, not merely decorating it.

TOK Are complex numbers discovered or invented? They began as a deliberate fiction, a symbol i defined into existence to make an equation solvable, which argues for invention. Yet once admitted, they turned out to describe electrical circuits, quantum states, and fluid flow with uncanny precision, as though they had been waiting in nature all along. When a tool built for pure convenience turns out to model the physical world so faithfully, what does that tell us about the relationship between mathematics and reality? Is the "unreasonable effectiveness" of complex numbers a coincidence, or a clue?

RW Complex numbers are the natural language of anything that oscillates. An alternating current, a radio wave, or a vibration is described by a rotating complex number whose real part is the physical signal; engineers add and scale these "phasors" to design circuits and filters that would otherwise need long trigonometric calculations. Quantum mechanics goes further still: the state of a particle is a complex-valued wave, and i sits at the centre of the equation that governs how it evolves. Signal processing in phones, impedance in power grids, and the mathematics of quantum computing all use the numbers of this chapter.

EXT De Moivre's theorem hints at a whole hidden unity. The identity $e^{i\theta} = \cos\theta + i\sin\theta$ says that the exponential function and the trigonometric functions, which look nothing alike on the real line, are the same function seen along different directions in the complex plane. Growth and oscillation, e^x and $\sin x$, are one thing viewed two ways. Push this further and you reach complex analysis, the study of functions of a complex variable. Such functions turn out to be far more tightly constrained than real ones, and problems about ordinary integers, such as how the prime numbers are distributed, are answered by studying functions over the complex plane.

End-of-Chapter Problems

1. Given $z = 3 + 4i$ and $w = 1 - 2i$:

(a) find $z + w$; [1]
(b) find zw; [2]
(c) find $\frac{z}{w}$; [2]
(d) find $|z|$. [1]

2. Express $\dfrac{2+i}{3-i}$ in the form $a + bi$. [4]

3. Find the modulus and the principal argument of

(a) $z = -1 + i$; [3]
(b) $z = \sqrt{3} - i$. [3]

4. Write $z = 1 + \sqrt{3}\,i$ in modulus–argument form and in Euler form. [4]

5. Use De Moivre's theorem to evaluate

(a) $(1 + \sqrt{3}\,i)^6$; [3]
(b) $(1 + i)^{10}$; [3]
(c) $\left(\dfrac{1+i}{\sqrt{2}}\right)^{12}$. [3]

6. Find the square roots of each number in the form $a + bi$.

(a) $3 + 4i$ [4]
(b) $-5 + 12i$ [4]

7. Given $z = 2\,\text{cis}\,\frac{\pi}{3}$:

(a) find z^2; [2]
(b) find z^3. [2]

8. The equation $z^2 + bz + c = 0$ has real coefficients and $1 - 2i$ as a root. Find b and c. [4]

9. Find the three cube roots of $8i$, giving each

(a) in modulus–argument form; [3]
(b) in exact Cartesian form. [3]

10. Simplify

(a) i^{2026}; [2]
(b) i^{-7}. [2]

11. Find all solutions of $z^3 = 27$, giving each in modulus–argument form. [4]

12. Find real numbers x and y such that $(x + yi)(2 - i) = 3 + i$. [4]

13.

(a) Express -8 in the form $re^{i\theta}$, with $-\pi < \theta \leq \pi$. [3]
(b) The complex number z has $|z| = 2$ and $\arg z = \frac{\pi}{6}$. Write z in Cartesian form. [3]

14.

(a) Find the four fourth roots of -16, giving each in the form $a + bi$. [5]
(b) Find all solutions of $z^4 = 1$, and show that they sum to zero. [4]

15. Solve $2z^2 + 2z + 5 = 0$. [4]

16. Given $z = \cos\theta + i\sin\theta$, show that $z + \dfrac{1}{z} = 2\cos\theta$. [4]

17. The points representing 1, i, -1 and $-i$ are plotted on an Argand diagram. State the shape they form and find its area. [3]

18. Given that $2+3i$ is a root of a polynomial with real coefficients, find a real quadratic factor of that polynomial. [4]

19. Solve $z^2 = 2i$, giving the roots in the form $a + bi$. [4]

20. Given $z = 2 + 2i$, find $|z|$ and $\arg z$, and hence write z in Euler form. [4]

21. Find the complex number z satisfying $2z - z^* = 3 + 12i$. [4]

22. Prove De Moivre's theorem, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, for all $n \in \mathbb{Z}^+$ by mathematical induction. [7]

23. Use De Moivre's theorem to show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$. [6]

24. Use De Moivre's theorem to show that $\tan 3\theta = \dfrac{3t - t^3}{1 - 3t^2}$, where $t = \tan\theta$. [7]

25. One cube root of a complex number w is $1+\sqrt{3}\,i$. Find w and the other two cube roots. [6]

26. In this question you will investigate the fifth roots of unity, the solutions of $z^5 = 1$.

(a) Write down the five roots in the form $\operatorname{cis}\theta$. [3]
(b) Show that the five roots sum to zero, using the sum of a geometric series. [3]
(c) Let $\omega = \operatorname{cis}\frac{2\pi}{5}$. Show that every fifth root of unity is a power of ω. [2]
(d) Deduce that $\omega + \omega^2 + \omega^3 + \omega^4 = -1$. [2]
(e) The five roots are plotted on an Argand diagram and joined to form a regular pentagon. Show that its area is $\frac{5}{2}\sin\frac{2\pi}{5}$, and evaluate this to three significant figures. [5]

27. Let $z = \operatorname{cis}\theta$.

(a) Show that $z^n + \dfrac{1}{z^n} = 2\cos n\theta$ for any $n \in \mathbb{Z}^+$. [3]
(b) Expand $\left(z + \dfrac{1}{z}\right)^4$ using the binomial theorem, and hence show that $16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$. [5]
(c) Deduce that $\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$. [2]
(d) Hence evaluate $\displaystyle\int_0^{\pi/2} \cos^4\theta \, d\theta$ exactly. [4]
(e) Explain what this method achieved that integrating $\cos^4\theta$ directly would not. [2]

28. The polynomial $P(z) = z^4 - 2z^3 + 14z^2 - 18z + 45$ has real coefficients, and $1 + 2i$ is a root.

(a) Write down another root of $P(z)$, and give a reason. [2]
(b) Find a real quadratic factor of $P(z)$. [3]
(c) Find the other quadratic factor. [3]
(d) Hence write down all four roots of $P(z)$. [3]
(e) The four roots are plotted on an Argand diagram. Describe the symmetry of the picture, and state the feature of $P(z)$ that guarantees it. [3]

Challenge Questions

29. **A sum of cosines from a seven-sided polygon.** Let $\omega = \operatorname{cis}\frac{2\pi}{7}$.

(a) Show that every solution of $z^7 = 1$ other than $z = 1$ satisfies $z^6+z^5+z^4+z^3+z^2+z+1 = 0$. [3]
(b) Show that $\omega^k + \omega^{7-k} = 2\cos\dfrac{2\pi k}{7}$ for $k = 1, 2, 3$. [3]
(c) Hence show that $\cos\dfrac{2\pi}{7} + \cos\dfrac{4\pi}{7} + \cos\dfrac{6\pi}{7} = -\dfrac{1}{2}$. [4]
(d) Verify this on your calculator. [2]
(e) Explain how the same argument applies to $z^{2m+1} = 1$, and state the general result. [3]

30. Rotation is multiplication. Throughout, points of the plane are identified with complex numbers.

(a) Explain why multiplying by $\operatorname{cis}\alpha$ rotates a point about the origin through α. [2]
(b) Show that rotating z about the point w through α gives $z' = w + (z - w)\operatorname{cis}\alpha$. [3]
(c) The triangle with vertices 0, 1 and z is equilateral. Find the two possible values of z. [4]
(d) Show that $\zeta = \operatorname{cis}\frac{\pi}{3}$ satisfies $\zeta^2 = \zeta - 1$. [2]
(e) A triangle has vertices a, b, c, labelled so that $c - a = (b - a)\zeta$. Show that $a^2 + b^2 + c^2 = ab + bc + ca$. [4]

31. How far does De Moivre reach? The syllabus states De Moivre's theorem for integer n and notes that it extends to rational exponents. This question examines what that extension has to say.

(a) Using the formula for nth roots, write down the three values of $(\operatorname{cis}\theta)^{1/3}$. [3]
(b) Explain why the statement $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ needs restating when n is not an integer. [3]
(c) Take $\theta = \pi$ and $n = \frac{1}{2}$. Evaluate $\cos n\theta + i\sin n\theta$, find both square roots of $\operatorname{cis}\pi$, and comment on what you find. [4]
(d) State a version of the theorem valid for rational $n = \frac{p}{q}$ in lowest terms, saying how many values the left-hand side takes. [3]